# Impulse Decisions Auction Portal – Developer Brief

Oct 7, 2026 · @James Marsden

## Overview

We need a secure, production-ready web app that replaces Impulse Decisions' paper promissory notes at in-person charity auctions with digital notes, e-signatures and Stripe payments. A working prototype already exists and defines the screens, wording and workflow; this brief covers what must be built to run it for real.

**The prototype.** A single HTML file (about 150 KB) built in Claude, supplied alongside this brief. It is a reference for behaviour, layout and copy, not production code: it has no server, its permissions are enforced only on screen, and payments, postcode lookup and email are simulated. The developer should rebuild it on a proper stack, matching the prototype screen for screen unless this brief says otherwise.

**What the developer is needed for:**

- A secure backend and database for events, notes, signatures and staff.
- Real sign-in with two-factor authentication, and roles enforced on the server.
- Live Stripe payments: online payment links and QR codes, plus Stripe Terminal card readers.
- Royal Mail postcode lookup, emailed receipts with a generated PDF, and SMS payment links.
- Hosting, backups, monitoring, GDPR compliance and ongoing support.

**Users.** Impulse Decisions staff at events (phones and tablets), event managers and master managers (laptops and tablets). Customers never log in; they sign on a staff device and pay on their own phone or a card reader.

## Users, roles and permissions

Three roles, enforced on the server for every request: a user's session must never return an event, note or report their role does not allow, whatever the interface shows.

| Capability | Master manager | Event manager (assigned events) | Team member |
|---|---|---|---|
| Events visible | All, any date | Events they manage, any date | Only events they are working, on the event date (until 06:00 next day) |
| Create events | Yes | No | No |
| Event setup: team, tables | Yes | Yes | No |
| Appoint event managers | Yes | No | No |
| Board totals, Report, CSV export | Yes | Yes | No |
| Create and sign notes, take payments | Yes | Yes | Yes |
| Amend a signed, unpaid note | Yes | Yes | Yes |
| Reassign collections, move all | Yes | Yes | Yes |
| Staff list: add | Yes | Yes | No |
| Staff list: remove | Yes | No | No |
| Settings (fees, delivery rates, links) | Yes | No | No |
| Add or remove master managers | Yes | No | No |

The account owner is always a master and cannot be removed; there must always be at least one master.

Masters can be added or removed at any time from Settings, taking effect immediately.

Staff without a login (casual or agency) can be listed and allocated tables; a logged-in colleague works their list.

Every permission change is written to the audit log (see Authentication and security).

# Authentication and security

Every login requires two-factor authentication; staff with Microsoft 365 accounts sign in with Microsoft, and everyone else uses an authenticator app.

**Microsoft sign-in (preferred).** "Sign in with Microsoft" via Microsoft Entra ID, with MFA enforced by Impulse Decisions' Microsoft 365 policy. Disabling a Microsoft account removes portal access.

**Portal login (casual staff).** Email and password plus a time-based authenticator code (TOTP, e.g. Microsoft or Google Authenticator). SMS codes only as a fallback.

**Masters.** Two-factor at every sign-in; no "remember this device".

**Shared tablets.** Automatic sign-out after 15 minutes of inactivity (configurable). The signing screen hides navigation so a customer cannot leave the note.

**Audit log.** Sign-ins, permission changes, note signing, amendments, payments, reassignments and exports, with user, time and IP address.

**Data protection (UK GDPR).** UK or EU hosting; encryption in transit and at rest; customer data restricted by role; a retention period for notes and signatures agreed with Impulse Decisions; export and deletion on request.

**Card data.** Never collected or stored by the portal. Stripe handles all card details, keeping the portal out of most PCI DSS scope. The paper form's card-details section is not reproduced.

**Signature evidence.** Store the signature image, timestamp, witness name, device details and a hash of the signed note so the record is tamper-evident, in line with UK electronic signature law.

# Functional requirements

The prototype is the reference for every screen below; behaviour, wording and validation should match it.

**Events and staff**

Masters create events (name, beneficiary, date) and appoint event managers per event.

Event setup: tick who is working tonight and give each person tables ("1-8, 12"); warn on clashes. Changeable at any time; removing someone moves their uncovered collections to Unassigned.

Staff list maintained by managers: add colleagues from the organisation directory, or by name only for staff without a login.

**Promissory notes**

Staff enter table number, customer details (name, mobile, email, Royal Mail postcode lookup, address) and lots (lot number, item, type: Experience, Memorabilia, Artwork, Charity, Other; bid amount).

Overseas delivery rows: destination from the rate list plus number of items; UK delivery free.

Optional service fee (currently 2.5% of lots plus delivery), included by default, removable by tick box. Rate and delivery prices editable by masters; signed notes keep the rates they were signed at.

Notes field for the team member or customer. Autosave plus explicit Save.

Customer signing page: details summary, lots, totals, delivery information, agreement wording, key bidder terms (clause 3), links and QR code to the full terms and ABTOT information, consent ticks (terms, GDPR, optional marketing), signature, witness name, and "taking the item(s) tonight?" with a collection signature if yes. "Back to note" returns to editing.

**Amendments**

A signed, unpaid note can be amended after confirmation. The original signed version is kept unchanged; a reason is required; the customer re-signs the amended version.

Removed lots are recorded as retracted (lot, amount, reason, time, user) and appear crossed out in the lot report. No retraction charges are added automatically.

Paid notes cannot be amended; corrections go through a refund (see Payments).

**Board and collections**

Board columns: Being prepared, Awaiting payment, Paid, Collected, with search. Totals visible to managers only.

Collections: each note is allocated by table, or reassigned to a named colleague. "My collections" list, Unassigned list, per-person totals, "Move all" for illness or busy staff. All changes appear on every device in real time.

**Reports**

By lot number (ranked highest bid first where a lot has several winners, retracted lots shown) and by promissory note. Totals by payment method and by team member. CSV export of each view.

**Customer email and PDF**

On signing and on payment, email the customer a branded message (formal tone) with the signed note as a two-page PDF attachment matching the prototype layout, plus a pay link if unpaid. Footer: full terms link, ABTOT statement, company address, registration and VAT numbers.

# Payments

All card payments go through Impulse Decisions' existing Stripe account; the server creates each payment for one note's exact total, tagged with the note ID, and only Stripe's confirmation marks a note paid.

**Payment methods**

| Method | How it works | Stripe product |
|---|---|---|
| Customer's phone | QR code on the staff device opens a payment page for that note; also sent by SMS and email | Stripe Checkout |
| Card reader | Staff pick the note and their reader; the amount appears on the reader; customer taps | Stripe Terminal, server-driven, smart readers (e.g. BBPOS WisePOS E or Stripe Reader S700) |
| Pay later | Link sent by SMS and email; automatic reminders until paid | Stripe Checkout |
| Cash, bank transfer, cheque, separate card machine | Recorded manually with an optional reference | None |

**Flow for a card payment**

1. Note signed; server stores it as Awaiting payment.
2. Server creates a Checkout Session or Terminal PaymentIntent with amount, currency GBP and note ID in metadata.
3. Customer pays on their phone or the reader. Card data goes directly to Stripe.
4. Stripe sends a webhook ( `checkout.session.completed` or `payment_intent.succeeded` ); the server verifies its signature, marks the note Paid, records method, staff member and time, and emails the receipt.
5. Every open device updates in real time.

**Requirements**

Webhooks are the source of truth and must be idempotent; a reconciliation job flags any payment without a matching note.

Card readers registered to a Stripe Terminal location, named "Reader 1" etc.; each staff member picks their reader.

A 4G router is recommended at venues with unreliable wifi.

Refunds for paid notes are issued from the portal by managers through Stripe, with a reason, and recorded against the note.
Payouts, fees and per-event totals must reconcile with Stripe reports.

## Integrations

Five external services replace what the prototype simulates; all are called from the server, never the browser.

| Service | Purpose | Notes |
|---|---|---|
| Royal Mail Postcode Address File | Postcode lookup that fills the address | Via a licensed provider such as Ideal Postcodes, Loqate or getAddress.io; per-lookup cost |
| Transactional email | Signed note, receipt, pay links, reminders | Sent from bookings@impulsedecisions.com with SPF, DKIM and DMARC set up; e.g. Postmark or Microsoft 365 |
| PDF generation | Two-page promissory note attached to emails and stored with the note | Server-rendered to match the prototype's "View PDF copy" layout |
| SMS | Payment links and reminders | UK provider such as Twilio; sender name "Impulse" if supported |
| Microsoft Entra ID | Staff sign-in and MFA | See Authentication and security |

## Data, hosting and quality

A relational database with real-time updates to connected devices; the prototype's data shapes are a starting point.

**Core records**

| Record | Key fields |
|---|---|
| Event | name, beneficiary, date, staff working, table allocations, event managers |
| Staff member | identity (Microsoft or portal login), name, has-login flag, master flag |

| Record | Key fields |
|---|---|
| Promissory note | event, reference (PN-XXXX), table, customer details, lots, delivery rows, fee rate and opt-out, notes, consents, signature, witness, collection signature, status, assignee, payment method, Stripe IDs, version |
| Note version | each previously signed version: lots, totals, signature, time, amendment reason |
| Retracted lot | lot, item, amount, reason, time, user |
| Settings | service fee %, delivery rates, terms and ABTOT links |
| Audit entry | user, action, record, time, IP |

**Hosting and non-functional**

- UK or EU cloud hosting with daily backups, 30-day retention and a tested restore.
- Separate test and live environments; Stripe test mode in test.
- Works on current iOS and Android phones, iPads and desktop browsers; phone layout with bottom navigation as in the prototype.
  Accessible to WCAG 2.2 AA.
- Pages respond in under 2 seconds on 4G; payment confirmations appear on all devices within 5 seconds.
  Error and uptime monitoring with alerts during events.

**Branding**

Follow the Impulse Decisions Brand Guidelines 2025, as already applied in the prototype: League Gothic uppercase headings, Gotham Book body text (licensed web font; Montserrat used in the prototype as a stand-in), Impulse Red #E62D2F, grey #525251, black, accent colours for status, the horizontal logo at no less than 20px high, formal brand voice in customer-facing text.

## Delivery, acceptance and open questions

Build in three phases so the team can use the core flow at a live event before every integration is finished.

1. **Core platform.** Sign-in with two-factor, roles enforced on the server, events, staff, notes, signing, amendments, board, collections, reports, audit log.

2. **Payments.** Stripe Checkout links and QR codes, webhooks, cash and other methods, refunds, then Stripe Terminal readers.
3. **Integrations and polish.** Royal Mail lookup, email with PDF, SMS, reminders, monitoring, accessibility review.

## Deliverables

- Source code in a repository owned by Impulse Decisions, with setup and deployment documentation.
- Test and live environments, configured with Impulse Decisions' Stripe, Microsoft and provider accounts.
- Automated tests covering permissions, payments and amendments.
- A short staff guide and a handover session.
- An agreed support and maintenance arrangement.

## Acceptance criteria

- A team member cannot retrieve any event they are not working tonight, even by calling the server directly.
- A note is only marked paid after a verified Stripe webhook; duplicate webhooks do not double-record.
- An amended note keeps its original signed version unchanged and requires a new signature.
- A complete event can run from note to receipt on a phone, a tablet and a laptop at the same time, with all devices staying in sync.
- The emailed PDF matches the prototype layout and branding.

## Open questions for Impulse Decisions

- Retention period for customer data and signatures.
- Whether casual staff get portal logins or work only through colleagues.
- Card reader quantity and model, and venue connectivity.
- Email and SMS providers, if there are existing contracts.
- Solicitor sign-off on the digital promissory note wording and the pre-ticked service fee.
- Budget, timeline and who hosts and supports the system after launch.